

Week 6 Progress Report: NLP Feature Phase, Y/N Vocab and ViC

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Summary

This report covers the literature summary promised after your last reading list, the design decision it fed into, two new real-content item generators built from it (Y/N Vocab and ViC), and the first results of testing them through the actual calibration pipeline, replicated across five seeds. The headline finding is not uniform: real features produced a reliable gain for ViC in every condition, but for Y/N Vocab only in Warm-start. Three targeted hypotheses to explain that asymmetry were tested and all three were ruled out. I am reporting the asymmetry as a real, replicated finding with the cause still open, rather than forcing an explanation that the evidence does not support. Full detail follows.

1. Literature summary

Went through the full reading list. Two points changed how the generators below were built.

Hand-crafted linguistic features are the field's dominant approach, not a fallback. AlKhazaey et al.'s (2024) systematic review of 55 studies found 54 percent of extracted features were hand-crafted linguistic features (frequency, length, n-gram statistics) versus 8 percent from contextual embeddings.

Discrimination is barely studied at all. Only two papers in or near this list attempt to predict discrimination from content features rather than difficulty alone: R2DE (Benedetto et al., 2020) and Byrd and Srivastava (2022). Both report discrimination as harder to predict than difficulty from the same features, the same asymmetry you flagged as a known IRT fact two weeks ago.

Three more results shaped specific choices: Stepanek et al. (2023) found an elastic net won for continuous difficulty prediction and a random forest won for classification, both competitive with human expert ratings. Hsu et al. (2018) found word embeddings improved over hand-crafted features specifically for vocabulary-type items. Ha et al. (2019), on a real high-stakes exam, reported their best combined-feature model only reached a correlation of 0.32 against true difficulty, a useful ceiling to keep in mind.

2. Design decision: ground truth

My last email left open whether the NLP phase should keep a self-assigned ground truth (known, but still our own choice) or try to approximate the real, unknown relationship the paper itself faces. Since your reply did not settle this directly, I went with what I think is the sensible middle path rather than wait: real content and real computed features, with the underlying relationship to difficulty still self-assigned but grounded in the direction the literature actually supports, and calibrated so that a single feature's correlation with true difficulty lands in a realistic published range rather than being artificially clean. This keeps the true-parameter diagnostics usable while actually fixing what last week's investigation identified as the real limitation: uninformative synthetic features.

3. Y/N Vocab item generator

Real English words drawn from a corpus-derived frequency list, filtered against an offline dictionary to remove proper nouns and named entities that the raw frequency data included. Fake words are generated with a

character-level Markov model trained on the same real vocabulary, a real simplification against the paper's RNN approach, disclosed as such.

Difficulty follows the lexical decision literature directly: for real words, difficulty rises as frequency falls; for fake words, difficulty rises as the word looks more plausible. Discrimination is left deliberately weak and mostly noise, not dressed up as literature-grounded when it is not.

Validation caught and fixed six real issues before this was trusted: proper nouns in the word list, an unrealistically strong single-feature correlation, an absolute difficulty scale far outside where test-taker ability actually sits, low-quality fake words at scale, a Windows compatibility problem, and fake words that were accidentally real but uncommon. Each was found by inspecting actual generated output, not assumed away.

4. ViC item generator

Real sentences from a public English text corpus, filtered down to well-formed, complete sentences, with one real content word per sentence masked. One limitation stated plainly: the sentence source is mid-twentieth-century text, so item content reads as somewhat dated compared to a real modern DET sentence, even though the features computed from it are unaffected by that.

Difficulty follows cloze-test literature: harder as the target word gets rarer, and harder as more competing real words fit the same revealed letters and length (a direct implementation of the conditional-probability feature the paper itself describes). Discrimination is treated the same deliberately-weak way as Y/N Vocab.

Validation found and fixed three real issues in the sentence pool and target selection: stray formatting artifacts left over from the source corpus, sentence fragments that were not actually complete sentences, and function words being incorrectly selected as fill-in-the-blank targets. Before any of that, the source corpus itself was chosen by direct comparison against an initial alternative that read as too dated, not assumed to be fine. Also confirmed, separately from the issues above, that the generator scales cleanly to the real DET item bank size.

5. Results: real pipeline, both item types

Both generators were run through the actual calibration pipeline, not a diagnostic shortcut, and evaluated with the same official metric compared against your Table 1 and Table 2 throughout this project. Single run first, then five-seed replication before trusting anything.

Y/N Vocab, 5-seed replication:

Split	Pearson mean	Synthetic baseline	Gain (mean)	Gain (min)	Gain (max)	Gap vs paper
Cold	0.621	0.619	+0.002	-0.035	+0.027	0.164
Jump 20	0.865	0.857	+0.008	-0.037	+0.033	0.071
Jump 40	0.904	0.898	+0.006	-0.020	+0.034	0.051
Jump 80	0.930	0.928	+0.002	-0.011	+0.013	0.041
Warm 05-08	0.968	0.965	+0.003	+0.000	+0.008	0.019
Warm 05-15	0.970	0.965	+0.005	+0.002	+0.008	0.018

"Gain (min)" is the number that actually matters: only positive for Warm-start. For Cold-start and all three Jump-start conditions, at least one of the five replications had the synthetic baseline outperforming the

real-feature run, so the positive mean gain in those rows is noise around zero, not a reliable effect. Real features helped, reliably, only for Warm-start recalibration.

ViC, 5-seed replication:

Split	Pearson mean	Synthetic baseline	Gain (mean)	Gain (min)	Gain (max)	Gap vs paper
Cold	0.708	0.572	+0.136	+0.076	+0.168	0.035
Jump 20	0.939	0.903	+0.036	+0.009	+0.052	0.015
Jump 40	0.957	0.927	+0.030	+0.017	+0.040	0.022
Jump 80	0.967	0.943	+0.024	+0.010	+0.030	0.022
Warm 05-08	0.982	0.963	+0.019	+0.014	+0.025	0.017
Warm 05-15	0.982	0.964	+0.018	+0.016	+0.024	0.016

Every condition has a positive "gain (min)" here, Cold-start most of all. Every one of the five replications beat the synthetic baseline in every one of the six conditions. This is a real, consistent, replicated effect, the opposite pattern from Y/N Vocab, where Cold-start was exactly the condition that did not hold up.

6. Investigating the asymmetry

Y/N Vocab's real-feature gain only survives replication where there is a lot of training data (Warm-start). ViC's survives everywhere, including Cold-start. I ran three targeted tests to explain this. All three came back negative, and I want to report that plainly rather than force a tidy story onto a result that does not support one.

Raw feature signal strength: already measured during generator validation, a single feature's correlation with true difficulty is close to identical for both item types (about 0.37 for each). Not the explanation.

Response volume: hypothesis was that Y/N Vocab's branching difficulty formula (one effect for real words, a different one for fake words) needs more data per branch than ViC's single smooth rule. Tested directly, holding operational item count fixed and sweeping training data volume from about 14,000 to 102,000 rows, 3 seeds per point. Both curves came back essentially flat across that entire range; the gap is present at the smallest volume and does not narrow at the largest. Not the explanation.

Model discoverability: maybe the tree ensemble cannot discover the branching rule from raw features alone, regardless of data. Tested by adding two explicit interaction features that spell the branching out directly, 3 seeds each. Recovery improved only slightly (mean Pearson -0.039 to +0.020), closing the gap to ViC's level from about 0.26 to about 0.20, not nearly closed. Not the explanation.

I do not have a fourth well-motivated hypothesis queued up.

Hypothesis	Verdict
Raw feature signal strength	Ruled out
Response volume ('needs more data')	Ruled out
Model discoverability of branching rule	Ruled out

My recommendation is to report the asymmetry as a real, replicated finding with the cause still open, rather than keep guessing at explanations that have not held up.

Next steps

- Given three specific explanations are now ruled out rather than one untested guess, my own read is that the more useful next step is extending the same real-feature approach to the full DET-scale item banks, now that both generators are validated at small scale.
- Open to your view on whether the asymmetry itself is worth a more structured investigation later, separate from the main feature-phase work.